March Report on FYP

Machine Learning Assisted Real-time Diagnosis on CO2 Discharge Process

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Problem Statement

- Nanosecond pulsed CO_2 discharge in water produces H_2O_2 — a key oxidant for water treatment
- Optical Emission Spectroscopy (OES) captures real-time plasma spectra (200–900 nm, 701 channels)
- **Goal:** Predict H_2O_2 yield rate from OES data and discharge parameters using machine learning && conduct analysis on features' contribution and model performance
- **Challenge:** Only 20 experimental samples available (small-sample regime)

Project Phases

The project is conducted in 4 iterative phases, each building on the previous one to enhance model performance and interpretability:

1. **Phase 1:** Baseline modeling with PCA-reduced OES features + discharge parameters
2. **Phase 2:** Hyperparameter tuning of nonlinear models (RF, MLP, CNN) using Bayesian optimization
3. **Phase 3:** Domain-knowledge-driven manual feature engineering (13 physics-based OES features)
4. **Phase 4:** Interpretability analysis and feature reduction via multi-model consensus and backward elimination

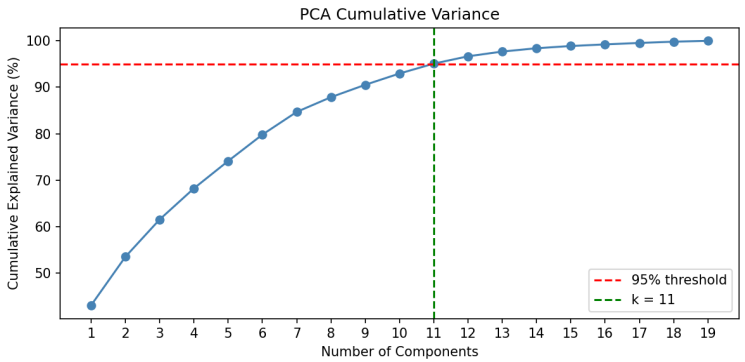
Experimental Data & Configurations

Config	Input Features	Dimensions
A	OES only (PCA-reduced)	11
B	Discharge parameters only	4
C	OES + discharge parameters	15 (P1) / 17 (P3)

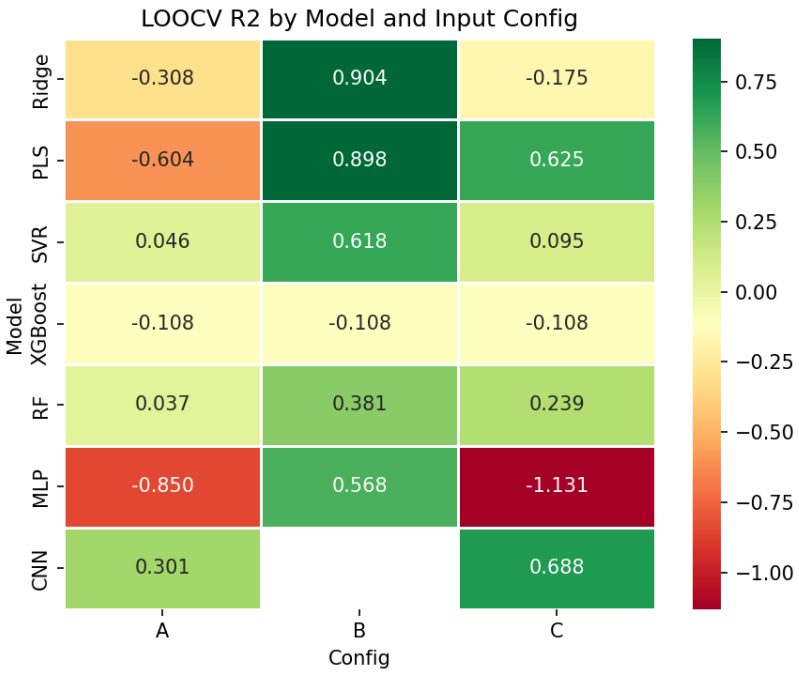
- **Discharge parameters:** frequency, pulse width, rise time, flow rate
- **Models tested:** Ridge, PLS, SVR, XGBoost, RF, MLP, CNN
- **Evaluation:** Leave-One-Out Cross-Validation (LOOCV) on 20 samples
- **Target:** H_2O_2 yield rate (range: 0.02–0.83)

Phase 1: PCA Dimensionality Reduction

- 701-channel OES spectrum reduced via PCA
- 11 principal components retained (95% cumulative variance)
- Severe curse of dimensionality: p/n ratio too high for 20 samples



Phase 1: Results — R^2 Heatmap (7 Models $\times$ 3 Configs)



- **Best:** Ridge Config B achieved $R^2 = 0.90$ (discharge params only)
- Discharge-parameter-only models (config B) outperformed OES-only (config A) and combined (config C) models

Phase 2: Motivation for Hyperparameter Tuning

- Phase 1 used default hyperparameters, which may not suit our small-sample, high-dimensional data
- Nonlinear models (RF, MLP, CNN) showed potential but **underperformed** due to either suboptimal settings or limited data
- Bayesian optimization can efficiently explore the hyperparameter space to enhance model performance. Thus can potentially recover nonlinear models' performance and better leverage OES features, especially in Config C

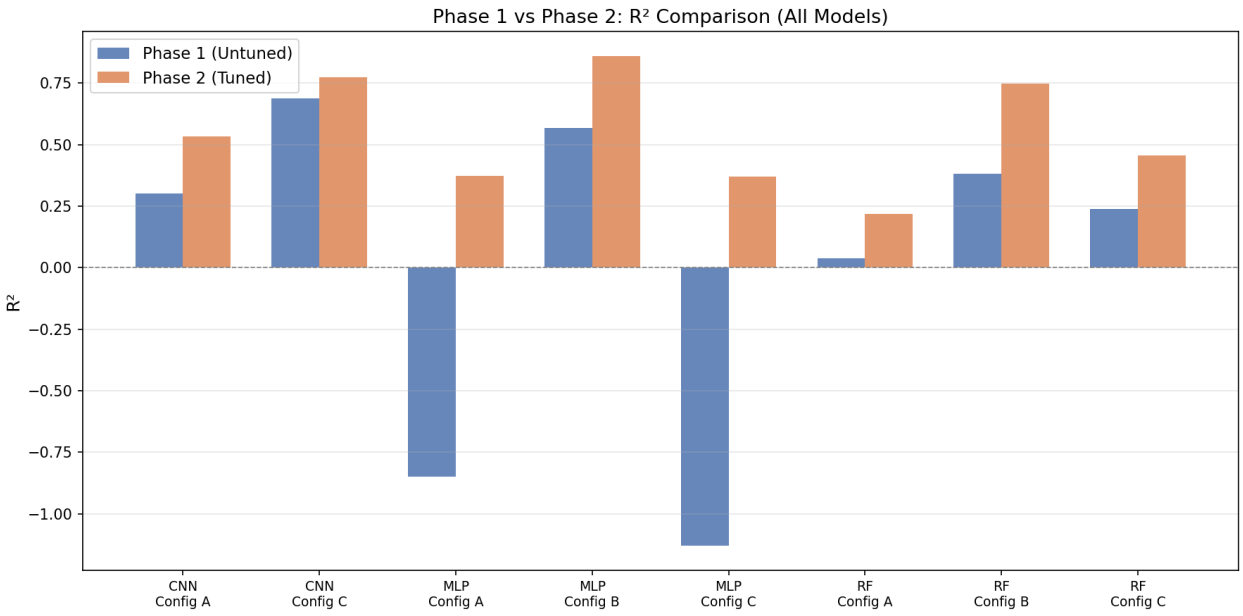
Phase 2: Bayesian Optimization

- Applied **Optuna** (TPE sampler) with nested LOOCV
- Tuned RF, MLP, and CNN: 100–200 trials per model-config pair
- Goal: recover nonlinear models from poor Phase 1 defaults

Key improvements after tuning:

Model	**Config**	**R^2 (P1)**	**R^2 (P2)**	**ΔR^2**
MLP	C	-1.13	0.37	$+1.50$
MLP	A	-0.85	0.37	$+1.22$
RF	B	0.38	0.75	$+0.37$
MLP	B	0.57	0.86	$+0.29$
CNN	C	0.69	0.77	$+0.09$

Phase 2: Results — Phase 1 vs Phase 2 Comparison



- Tuning universally beneficial, improving the non-linear ML models (MLP, RF, CNN) across all configs
- But Config B still dominant, and OES features via PCA remained weak even with optimized models

Phase 3: Motivation for Domain-Knowledge Feature Engineering

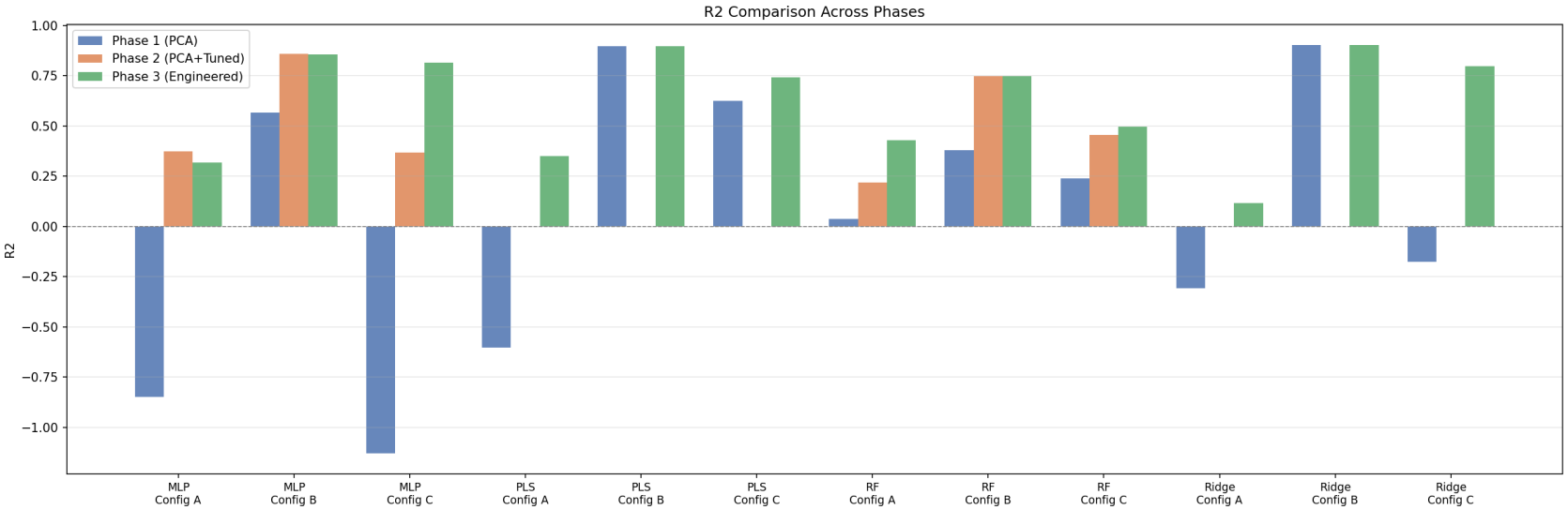
- PCA features are abstract, it only shows variance but lacks physical meaning, and may not capture the underlying plasma chemistry relevant to H_2O_2 production
- Leveraging domain knowledge (plasma chemistry) to engineer specific OES features (e.g., intensities of key emission lines, band integrals, ratios) can create more meaningful inputs that directly relate to chemical processes
- Hypothesis: Higher representative Physics-based features will enhance model interpretability and predictive power, especially in Config C where OES data is combined with discharge parameters

Phase 3: 13 Physics-Based OES Features

Replaced blind PCA with features grounded in plasma chemistry:

ID	Feature	Type	Species	Role
F1	I(309 nm)	Intensity	OH	Direct H_2O_2 precursor
F2	I(777 nm)	Intensity	O atom	CO_2 dissociation product
F3	I(656 nm)	Intensity	Hα	H_2O dissociation degree
F4	I(486 nm)	Intensity	Hβ	Electron diagnostics
F5	I(337 nm)	Intensity	N_2 SPS	High-energy e^- marker
F6	I(406 nm)	Intensity	CO_2^+	CO_2 ionization degree
F7	I(516 nm)	Intensity	C_2	Deep decomposition
F8	$\int$(306–312)	Band integral	OH 0-0 band	Robust OH emission
F9	$\int$(398–412)	Band integral	CO_2^+ FDB	Aggregate ionization
F10	$\int$(460–500)	Band integral	CO + Hβ	CO + H_2O dissociation
F11	I(309)/I(656)	Ratio	OH/Hα	OH availability
F12	I(777)/I(309)	Ratio	O/OH	Radical pool balance
F13	I(656)/I(486)	Ratio	Hα/Hβ	Balmer decrement

Phase 3: Significant Results — Cross-Phase Comparison

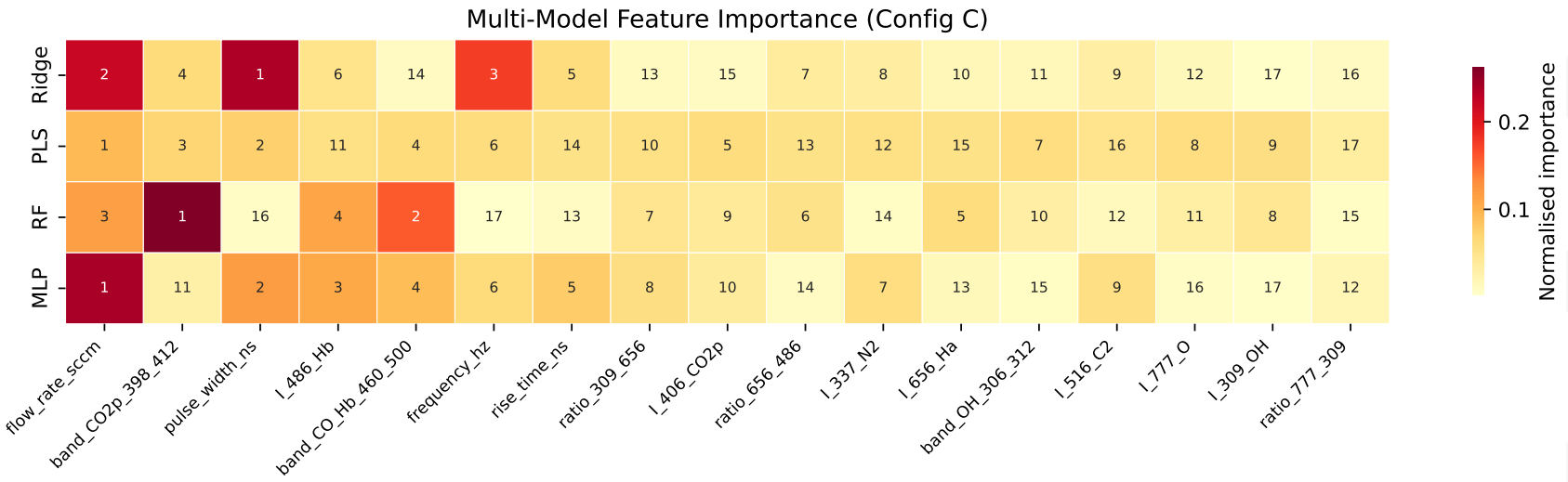


- **Ridge Config C:** R^2 improved from -0.17 to 0.80 ($+0.97$)
- **MLP Config C:** R^2 improved from -1.13 to 0.82 ($+1.95$)
- Domain knowledge dramatically unlocked OES predictive power

Phase 4: Motivation for Interpretability and Feature Reduction

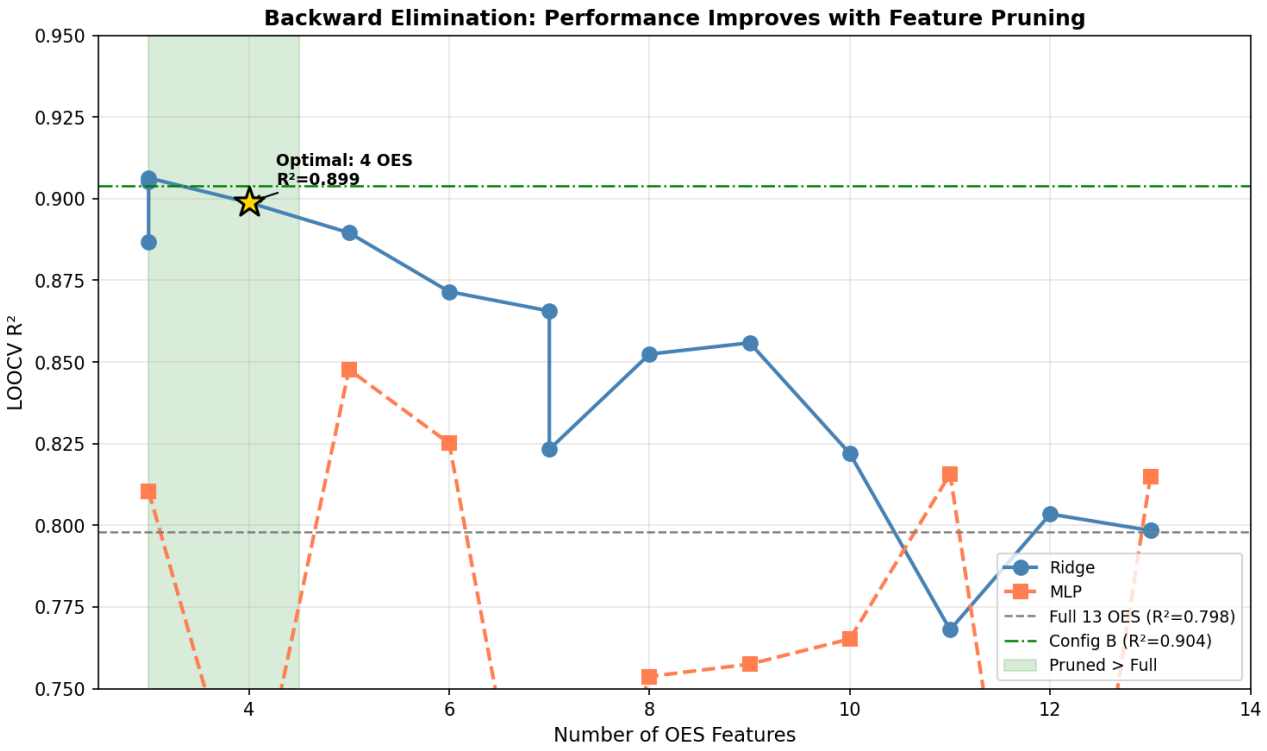
- Phase 3's physics-based features improved performance but increased model complexity (17 features), risking overfitting in our small-sample regime
- Understanding which features are most influential can enhance interpretability, avoid overfitting and probably improve generalization by removing redundant or noisy features (feature work)
- Hypothesis: A smaller subset of the most important features (e.g., top 5-7) can achieve comparable or better performance than the full set by reducing collinearity and noise, while also providing clearer insights into the key plasma processes driving H_2O_2 production

Phase 4: Multi-Model Feature Importance Consensus



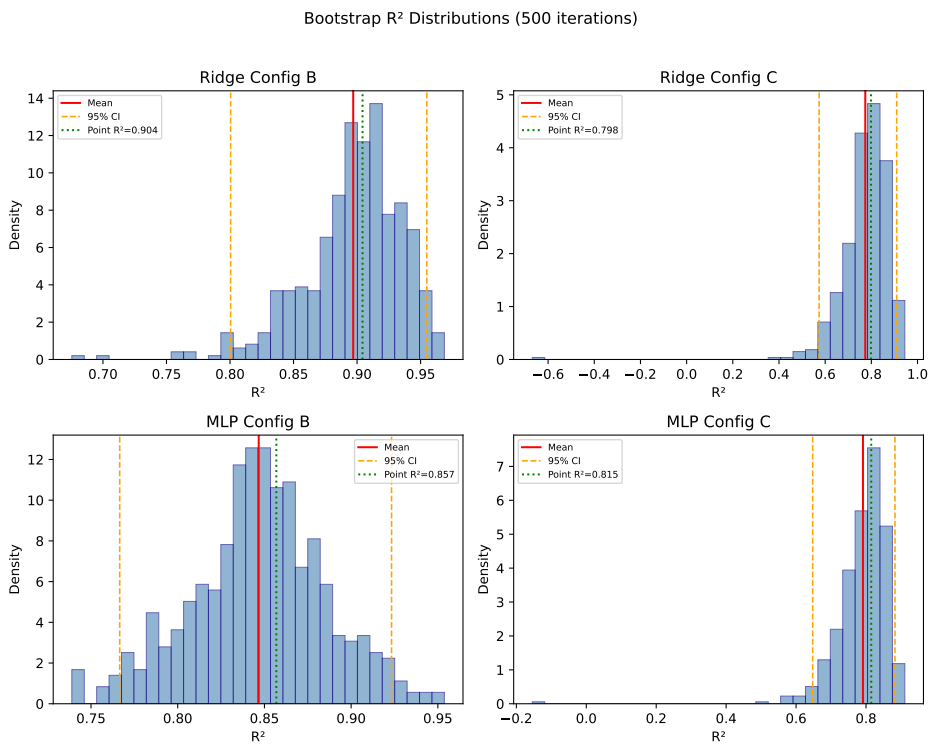
- Ranked 17 features across Ridge, PLS, RF, and MLP (SHAP)
- **Top/Significant features:** flow rate, CO_2^+ band integral, pulse width

Phase 4: Backward Elimination — Ablation Trajectory



- Removing redundant OES features **improves** R^2 (collinearity reduction)
- Optimal: **4 OES + 4 discharge params** $\rightarrow$ $R^2 = 0.899$
- Category ablation: 3 ratios alone + 4 params $\rightarrow$ $R^2 = 0.920$

Phase 4: Statistical Validation



Bootstrap 95% CIs (500 resamples):

Model	**R^2 [95% CI]**
Ridge B	0.90 [0.80, 0.95]
Ridge C	0.80 [0.57, 0.91]
MLP C	0.82 [0.65, 0.88]
MLP B	0.86 [0.77, 0.92]

Permutation test (2000 iter.):

- Observed $R^2 = 0.920$
- $p < 0.0005$

Phase 4: Final Optimized Model

Property	**Value**
Model	Ridge Regression
Total features	7
OES features (3)	I(309)/I(656), I(777)/I(309), I(656)/I(486)
Discharge params (4)	frequency, pulse width, rise time, flow rate
R^2 (LOOCV)	**0.920**
Permutation *p*-value	< 0.0005

- 3 OES intensity ratios + 4 discharge parameters = **7 features total**
- Outperforms the full 17-feature model (R^2 = 0.798) by removing collinear noise
- Simple, interpretable, and statistically significant

Progress Summary: R^2 Evolution Across Phases

Phase	Best Model	Config	R^2	Key Advancement
1	Ridge	B	0.904	Baseline; PCA for OES failed
2	MLP	B	0.861	Bayesian hyperparameter tuning
3	MLP	C	0.815	Domain-knowledge OES features
4	Ridge	C (pruned)	**0.920**	Feature reduction + interpretability

Key takeaways:

- Domain knowledge + feature reduction outperformed complex models on small data
- OES features became useful only after physics-guided engineering (Phase 3)
- Final model: simple Ridge with 7 features achieves $R^2 = 0.920$ ($p < 0.0005$)

Reference I

Thank You !